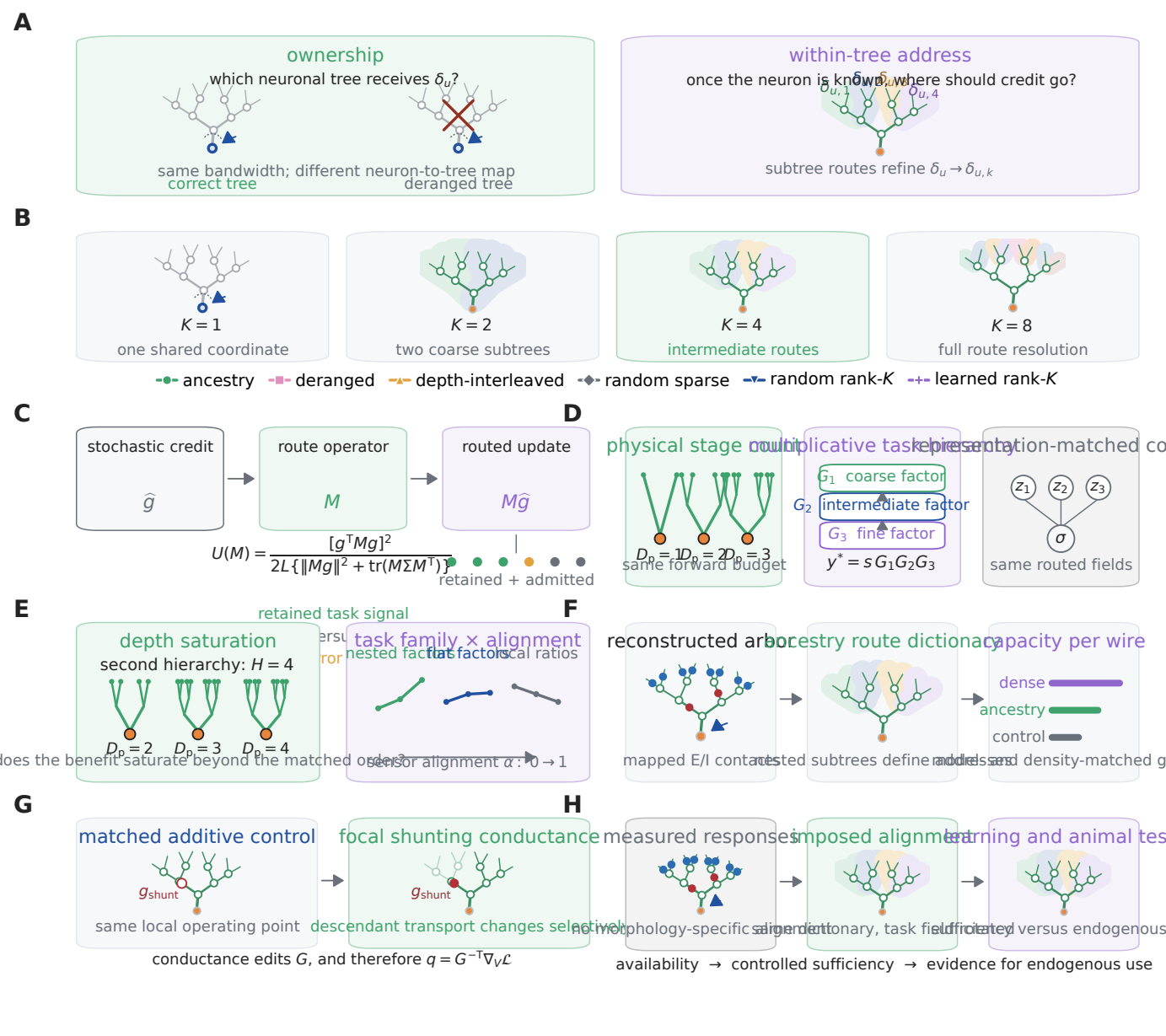

A
ownership
which neuronal tree receives δ_u?
same bandwidth; different neuron-to-tree map
correct tree
deranged tree
within-tree address
once the neuron is known, where should credit go?
$\delta_{u,1}$
$\delta_{u,2}$
$\delta_{u,3}$
$\delta_{u,4}$
subtree routes refine $\delta_u \to \delta_{u,k}$
B
$K = 1$
one shared coordinate
$K = 2$
two coarse subtrees
$K = 4$
intermediate routes
$K = 8$
full route resolution
ancestry
deranged
depth-interleaved
random sparse
random rank-K
learned rank-K
C
stochastic credit
$\widehat{g}$
route operator
M
routed update
$M\widehat{g}$
$U(M) = \frac{[g^\top Mg]^2}{2L\{\|Mg\|^2 + \mathrm{tr}(M\Sigma M^\top)\}}$
retained + admitted
D
physical stage count
multiplicative task representation
G_1 coarse factor
G_2 intermediate factor
G_3 fine factor
$y^* = s\,G_1G_2G_3$
$D_p = 1$
$D_p = 2$
$D_p = 3$
same forward budget
same routed fields
E
retained task signal
depth saturation
second hierarchy: $H = 4$
$D_p = 2$
$D_p = 3$
$D_p = 4$
does the benefit saturate beyond the matched order?
task family × alignment
nested factors
flat factors
local ratios
sensor alignment $\alpha : 0 \to 1$
F
reconstructed arbor
ancestry route dictionary
capacity per wire
dense
ancestry
control
mapped E/I contacts
nested subtrees define modules
address- and density-matched g
G
matched additive control
g_{shunt}
same local operating point
focal shunting conductance
g_{shunt}
descendant transport changes selectively
conductance edits G, and therefore $q = G^{-\top}\nabla_V\mathcal{L}$
H
measured responses
imposed alignment
learning and animal tes
no morphology-specific alignment
dictionary, task field
sufficient
versus endogenous
availability → controlled sufficiency → evidence for endogenous use